



**UG Program in Electronics and Communication
(Advanced Communication Technology)**

Experiment No. 8

Generation and Detection of Quadrature Phase Shift Keying (QPSK)

8.1 Aim: To study the Quadrature Phase Shift Keying (QPSK) generation and detection technique using MATLAB Simulink and plot the modulated and demodulated waveforms.

8.2 Course Outcome: Compare the performances of different digital modulation techniques

8.3 Learning Objectives:

- Understand the principles of QPSK modulation and demodulation techniques.
- Generate and detect QPSK signals and plot the modulated and demodulated waveforms.
- Explore noise impact using an AWGN (Additive White Gaussian Noise) channel in MATLAB Simulink.

8.4 Related Theory:

QPSK is a digital modulation scheme that conveys data by changing the phase of the carrier wave. Unlike BPSK, which shifts the phase by 180°, QPSK shifts the phase by 90°. QPSK transmits two bits of data per symbol, thus offering twice the data rate of BPSK within the same bandwidth.

QPSK can be visualized using a signal constellation diagram, where each point represents a unique symbol with four possible phase shifts: 0°, 90°, 180°, and 270°. This results in improved bandwidth efficiency.

QPSK Transmitter and Receiver:

1. Digital Data Generator: Generates the input binary data.
2. Random Integer Generator: Produces a range of integer values that represent the binary input.

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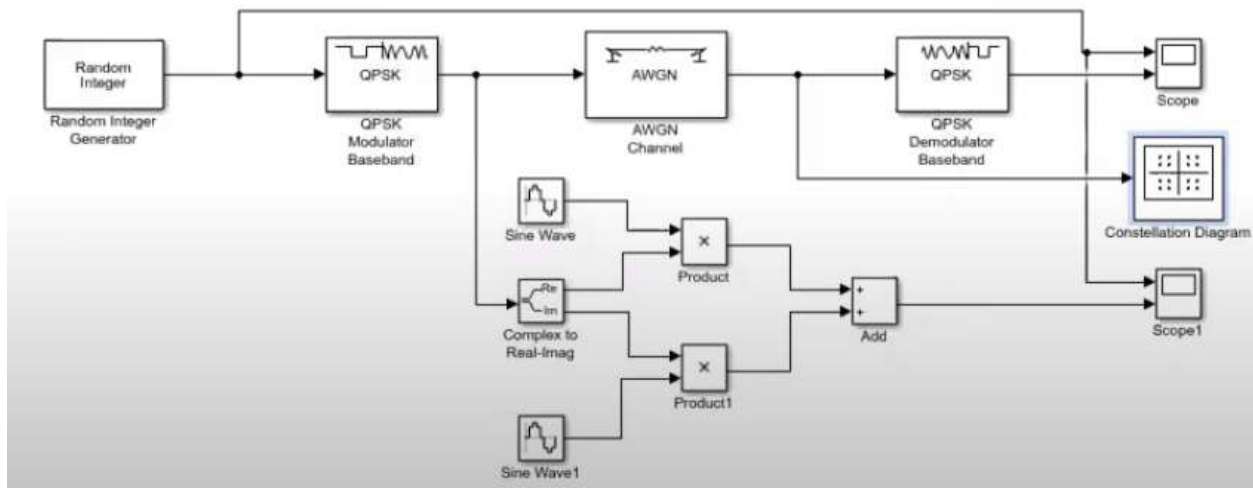
3. QPSK Modulator: Modulates the input data into phase-shifted carrier signals.
4. AWGN Channel: Introduces noise to simulate real-world transmission conditions.
5. QPSK Demodulator: Recovers the original data from the received noisy signal.

MATLAB Simulink Blocks:

1. Random Integer Generator: Generates the digital data stream as integers (0, 1, 2, 3) to be modulated.
2. QPSK Modulator Baseband: Modulates the integer stream into QPSK signals.
3. AWGN Channel: Simulates noise in the channel, affecting the signal's quality.
4. QPSK Demodulator Baseband: Demodulates the received noisy QPSK signal.
5. Scope: Visualizes the input data, modulated signal, and demodulated data.

Circuit Diagram:

QPSK Circuit Diagram in Simulink



Procedure:

1. Open MATLAB Simulink and create a new model.



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2. Add the following blocks:

- Ranger Integer Generator (from the Sources block library).
- QPSK Modulator Baseband (from the Digital Modulation block library).
- AWGN Channel (from the Channels block library).
- QPSK Demodulator Baseband (from the Digital Demodulation block library).
- Scope (from the Sinks block library) to visualize the signals.
- Constellation Diagram Block (from the Sinks block library) to plot the QPSK constellation.

3. Configure the blocks:

- Ranger Integer Generator: Set to output integers (0, 1, 2, 3) representing the 2-bit binary data.
- QPSK Modulator Baseband: Set to Gray coding.
- AWGN Channel: Set the SNR to a desired value
- QPSK Demodulator Baseband: Set to match the modulation parameters.
- Constellation Diagram Block: Connect the output of both the QPSK Modulator and QPSK Demodulator to display the transmitted and received constellations.

4. Run the simulation:

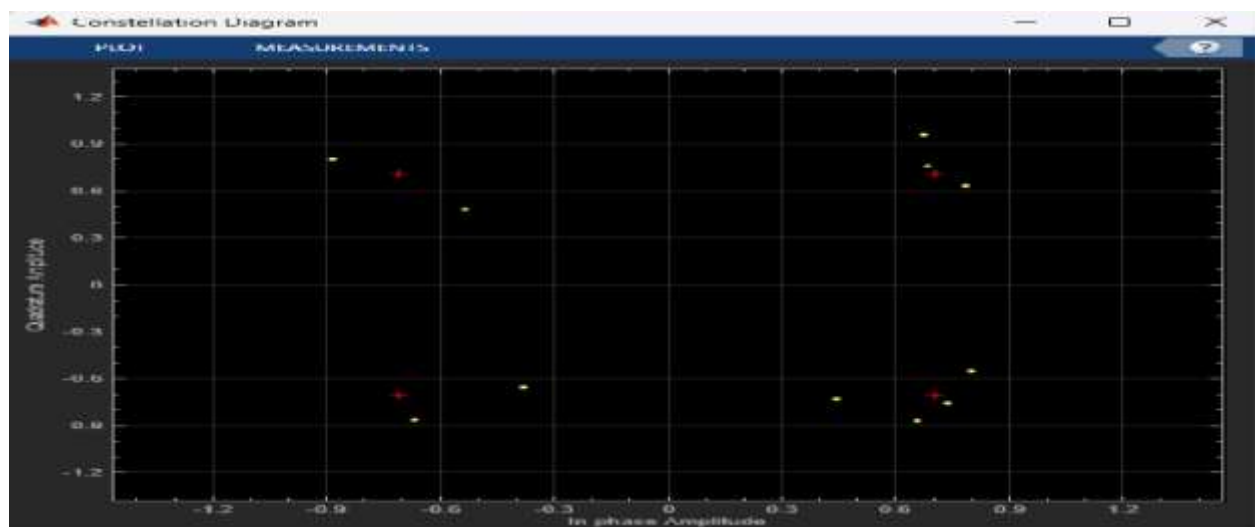
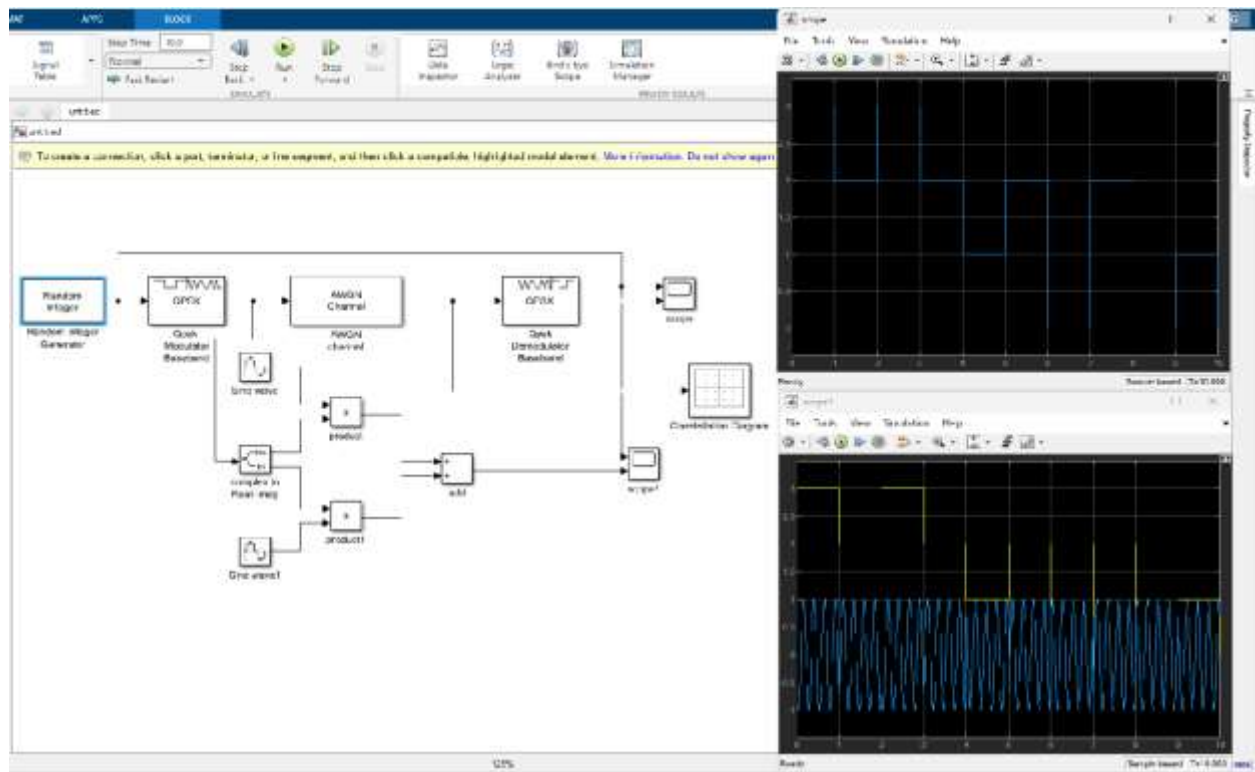
- Observe the waveforms on the Scope.
- View the constellation of the transmitted and received QPSK symbols on the Constellation Diagram.
- Analyze how noise affects the constellation points in the AWGN Channel.



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8.5 Output:

Simulation Circuit & Waveform & Plotted Graph





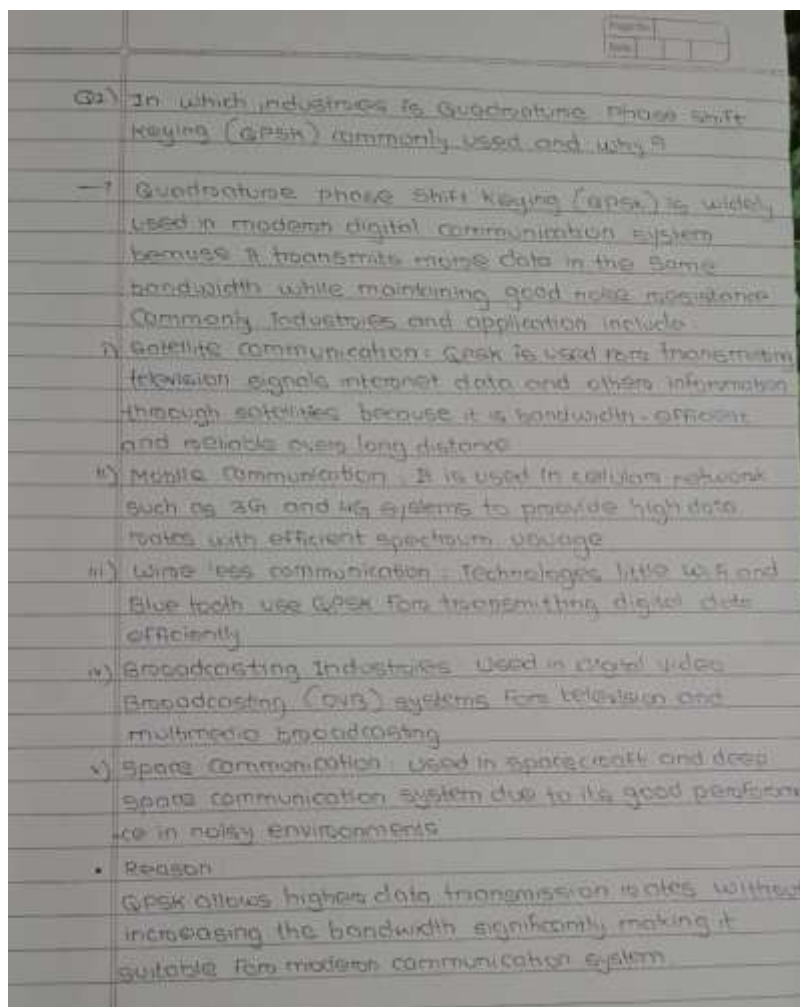
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8.6 Conclusion:

The QPSK generation and detection were successfully studied using MATLAB and Simulink. The modulated and demodulated waveforms were plotted and observed clearly. This experiment shows how QPSK can transmit digital data efficiently and recover the original signal at the receiver.

8.7 Questions:

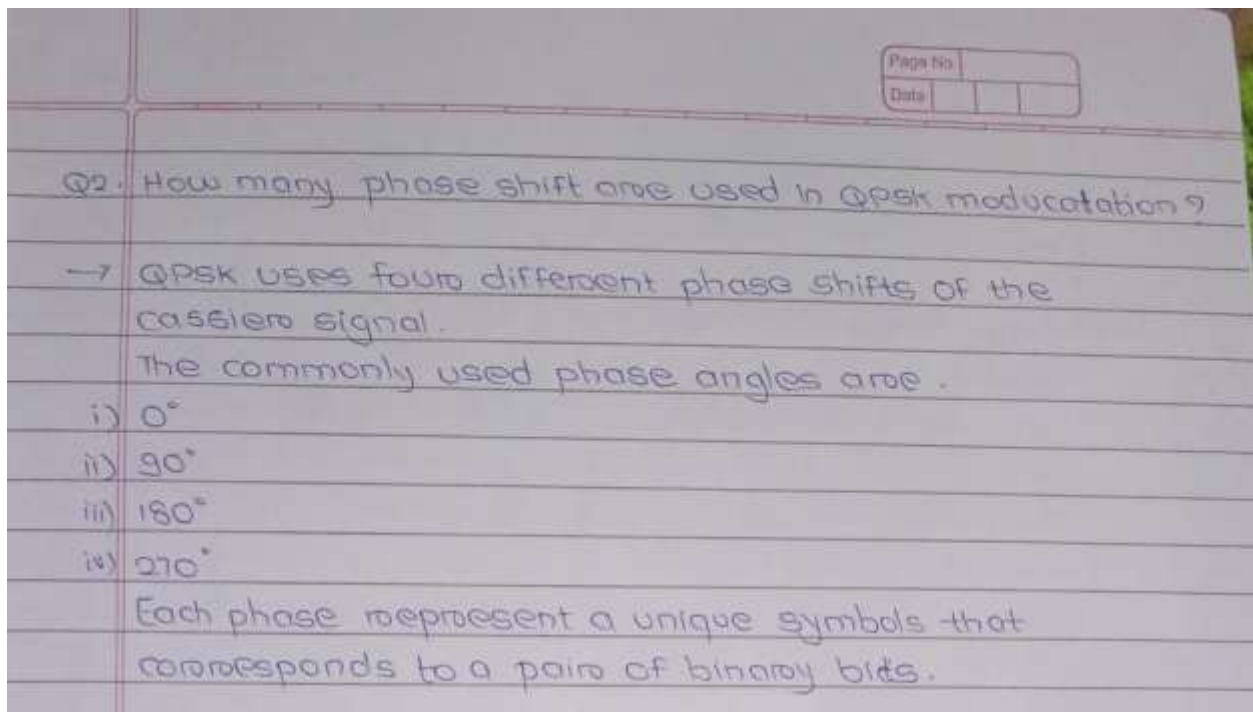
1 In which industries is Quadrature Phase Shift Keying (QPSK) commonly used, and why?





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2 How many phase shifts are used in QPSK modulation?





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3 How many bits are represented by each QPSK symbol?

Q3 How many bits are represented by each QPSK symbol?

→ In QPSK each symbol represented 2 bits of information

- Example mapping :

Bit	phase
00	0°
01	90°
10	180°
11	270°

This means two bits are transmitted in one symbol which increases the data rate compared to BPSK.



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4 What is the advantage of QPSK over BPSK?

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 Q4. What is the advantage of QPSK over BPSK ?
 -> The main advantage of QPSK over BPSK is higher data transmission efficiency.
 Advantages :
 i) Higher Data Rate :
 • BPSK transmits 1 bit per symbol.
 • QPSK transmits 2 bit per symbol.
 ii) Better Bandwidth Efficiency :
 • QPSK can transmit twice the data within the same bandwidth compared to BPSK.
 iii) Efficient spectrum usage :
 • It allows communication system to use limited frequency spectrum more effectively.
 iv) Suitable for high speed communication :
 • Because it carries more bits per symbol QPSK is ideal for modern digital communication system.